Solution Requirements

Date	30 September 2026
Team ID	[Your Team ID]
Project Name	Streamline IT Procurement: Automating Standard Laptop Orders with Workflow Automation
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (*What the system should do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (*How the system should be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	**Authentication**	Employees and IT managers must log in using Single Sign-On (SSO) integration with Azure AD/Okta to ensure secure identity verification.	**High**
2	**Authorization levels**	Role-Based Access Control (RBAC) defining standard employees (requestors), department managers (approvers), and IT administrators (procurement officers).	**High**
3	**External interfaces**	Integrates with ERP/ITSM solutions (e.g., ServiceNow, SAP) and vendor procurement APIs (e.g., Dell, Lenovo) for automated order placement.	**Medium**
4	**Transactions processing**	Automates workflow routing for laptop catalog selection, multi-level manager approval, order tracking, and inventory status updates.	**High**
5	**Reporting**	Provides dashboards and reports for order fulfillment time, total procurement costs, pending approvals, and inventory stock levels.	**Medium**
6	**Business rules, etc.**	Orders exceeding predefined budget thresholds require secondary finance approval; standard models auto-approve if under budget limit.	**High**
7	**Compliance to laws or regulations**	Ensures strict compliance with corporate procurement policies, SOX audit trails, and data protection regulations (GDPR/ISO27001).	**High**
8	**Other**	Automated email/Slack notifications sent to users at each stage (Submitted, Approved, Shipped, Delivered).	**Low**

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	**Performance & Speed**	Fast system response for catalog browsing, request submission, and automated workflow state transitions.	Page loads in < 2 seconds; automated approval routing triggers in < 5 seconds.
2	**Scalability**	System capable of handling organizational expansion and peak procurement cycles during company-wide refreshes.	Supports up to 10,000 concurrent users and 500 daily procurement requests without lag.
3	**Security & Data Privacy**	Protection of sensitive procurement records, employee personal details, and vendor payment data.	End-to-end AES-256 encryption for data at rest and TLS 1.3 for data in transit.
4	**Reliability & Availability**	High availability to ensure continuous ordering capabilities across global offices and time zones.	99.9% uptime per month with automated failover and daily database backups.
5	**Usability & Accessibility**	Intuitive user interface for easy laptop selection and clear status tracking for non-technical users.	Complies with WCAG 2.1 AA standards; zero user training required for basic orders.
6	**Other**	Maintainability and modularity to allow seamless integration of new laptop hardware models or vendors.	New laptop models or hardware bundles can be added to the catalog in < 15 minutes.